```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set(style="whitegrid")

df = pd.read_csv("/content/diabetic_data.csv")
df.head()
```

	encounter_id	patient_nbr	race	gender	age	weight	admission_type_id	discharge_disposition_id	admission_source_
0	2278392	8222157	Caucasian	Female	[0-10)	?	6	25	
1	149190	55629189	Caucasian	Female	[10-20)	?	1	1	
2	64410	86047875	AfricanAmerican	Female	[20-30)	?	1	1	
3	500364	82442376	Caucasian	Male	[30-40)	?	1	1	
4	16680	42519267	Caucasian	Male	[40-50)	?	1	1	

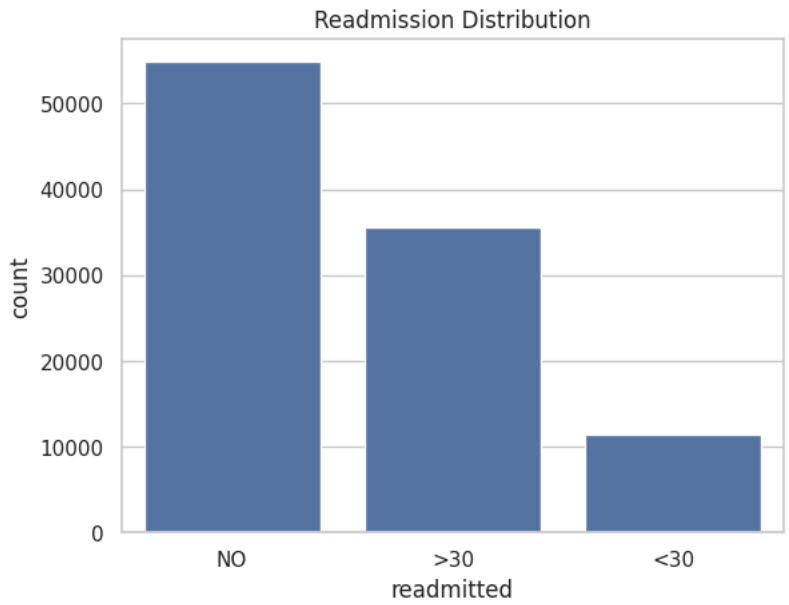
5 rows × 50 columns

```
df.shape
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 101766 entries, 0 to 101765
Data columns (total 50 columns):
 #   Column                    Non-Null Count   Dtype
---  ------                    --------------   -----
 0   encounter_id              101766 non-null  int64
 1   patient_nbr               101766 non-null  int64
 2   race                      101766 non-null  object
 3   gender                    101766 non-null  object
 4   age                       101766 non-null  object
 5   weight                    101766 non-null  object
 6   admission_type_id         101766 non-null  int64
 7   discharge_disposition_id  101766 non-null  int64
 8   admission_source_id       101766 non-null  int64
 9   time_in_hospital          101766 non-null  int64
 10  payer_code                101766 non-null  object
 11  medical_specialty         101766 non-null  object
 12  num_lab_procedures        101766 non-null  int64
 13  num_procedures            101766 non-null  int64
 14  num_medications           101766 non-null  int64
 15  number_outpatient         101766 non-null  int64
 16  number_emergency          101766 non-null  int64
 17  number_inpatient          101766 non-null  int64
 18  diag_1                    101766 non-null  object
 19  diag_2                    101766 non-null  object
 20  diag_3                    101766 non-null  object
 21  number_diagnoses          101766 non-null  int64
 22  max_glu_serum             5346 non-null    object
 23  A1Cresult                 17018 non-null   object
 24  metformin                 101766 non-null  object
 25  repaglinide               101766 non-null  object
 26  nateglinide               101766 non-null  object
 27  chlorpropamide            101766 non-null  object
 28  glimepiride               101766 non-null  object
 29  acetohexamide             101766 non-null  object
 30  glipizide                 101766 non-null  object
 31  glyburide                 101766 non-null  object
 32  tolbutamide               101766 non-null  object
 33  pioglitazone              101766 non-null  object
 34  rosiglitazone             101766 non-null  object
 35  acarbose                  101766 non-null  object
 36  miglitol                  101766 non-null  object
 37  troglitazone              101766 non-null  object
 38  tolazamide                101766 non-null  object
 39  examide                   101766 non-null  object
 40  citoglipton               101766 non-null  object
 41  insulin                   101766 non-null  object
 42  glyburide-metformin       101766 non-null  object
 43  glipizide-metformin       101766 non-null  object
 44  glimepiride-pioglitazone  101766 non-null  object
 45  metformin-rosiglitazone   101766 non-null  object
 46  metformin-pioglitazone    101766 non-null  object
 47  change                    101766 non-null  object
 48  diabetesMed               101766 non-null  object
 49  readmitted                101766 non-null  object
dtypes: int64(13), object(37)
memory usage: 38.8+ MB
```

	encounter_id	patient_nbr	admission_type_id	discharge_disposition_id	admission_source_id	time_in_hospital	num_lab_pr
count	1.017660e+05	1.017660e+05	101766.000000	101766.000000	101766.000000	101766.000000	1017
mean	1.652016e+08	5.433040e+07	2.024006	3.715642	5.754437	4.395987	
std	1.026403e+08	3.869636e+07	1.445403	5.280166	4.064081	2.985108	
min	1.252200e+04	1.350000e+02	1.000000	1.000000	1.000000	1.000000	
25%	8.496119e+07	2.341322e+07	1.000000	1.000000	1.000000	2.000000	
50%	1.523890e+08	4.550514e+07	1.000000	1.000000	7.000000	4.000000	
75%	2.302709e+08	8.754595e+07	3.000000	4.000000	7.000000	6.000000	
max	4.438672e+08	1.895026e+08	8.000000	28.000000	25.000000	14.000000	1

```
sns.countplot(x='readmitted', data=df)
plt.title("Readmission Distribution")
plt.show()
```

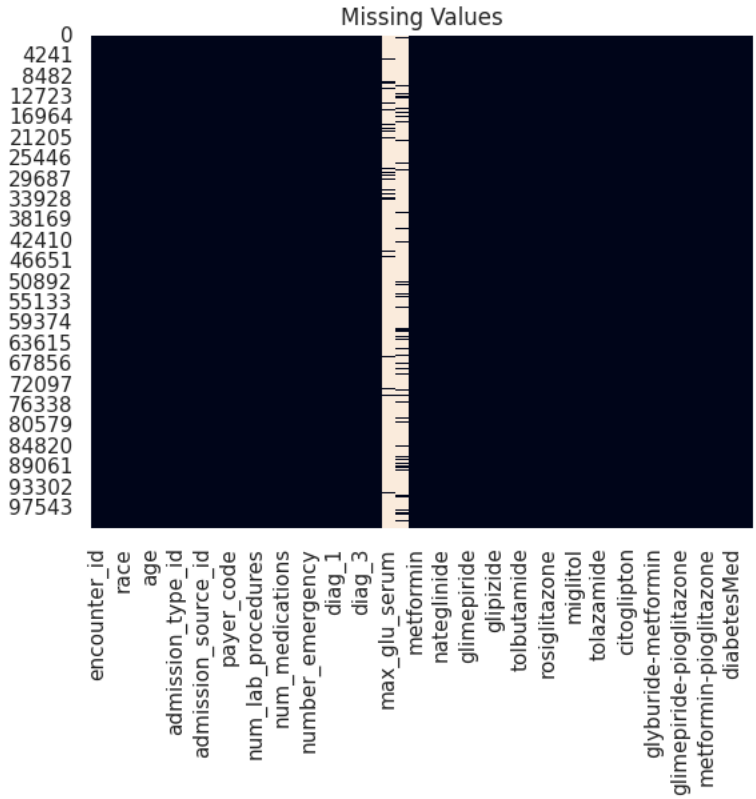



```
missing = df.isnull().sum().sort_values(ascending=False)
missing[missing > 0]
```

	0
max_glu_serum	96420
A1Cresult	84748

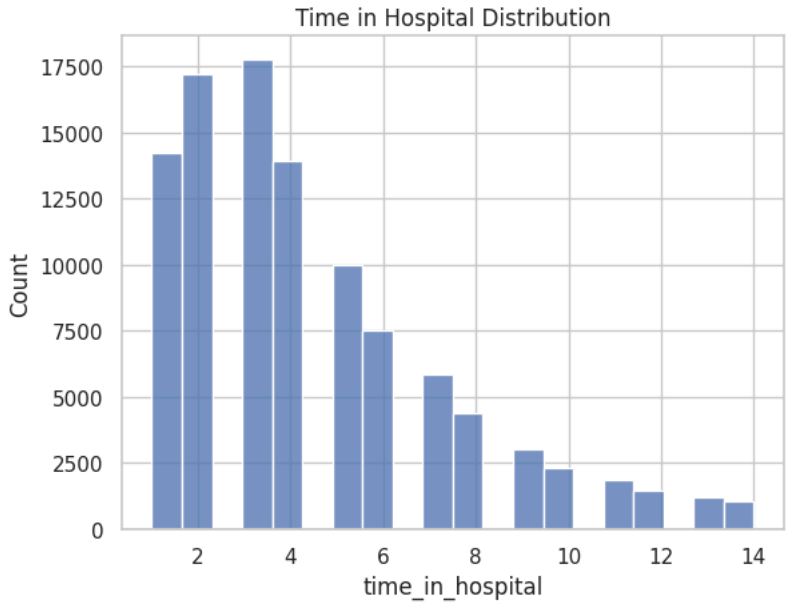
dtype: int64

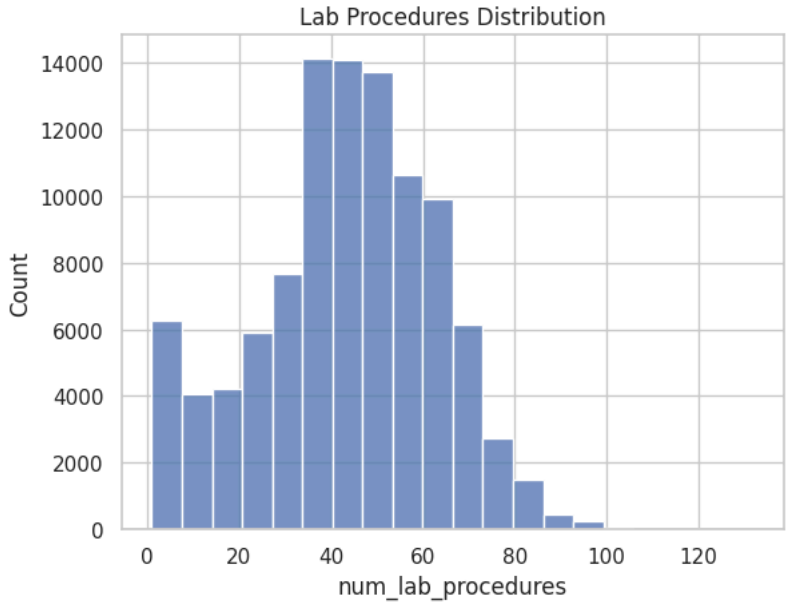
```
sns.heatmap(df.isnull(), cbar=False)
plt.title("Missing Values")
plt.show()
```



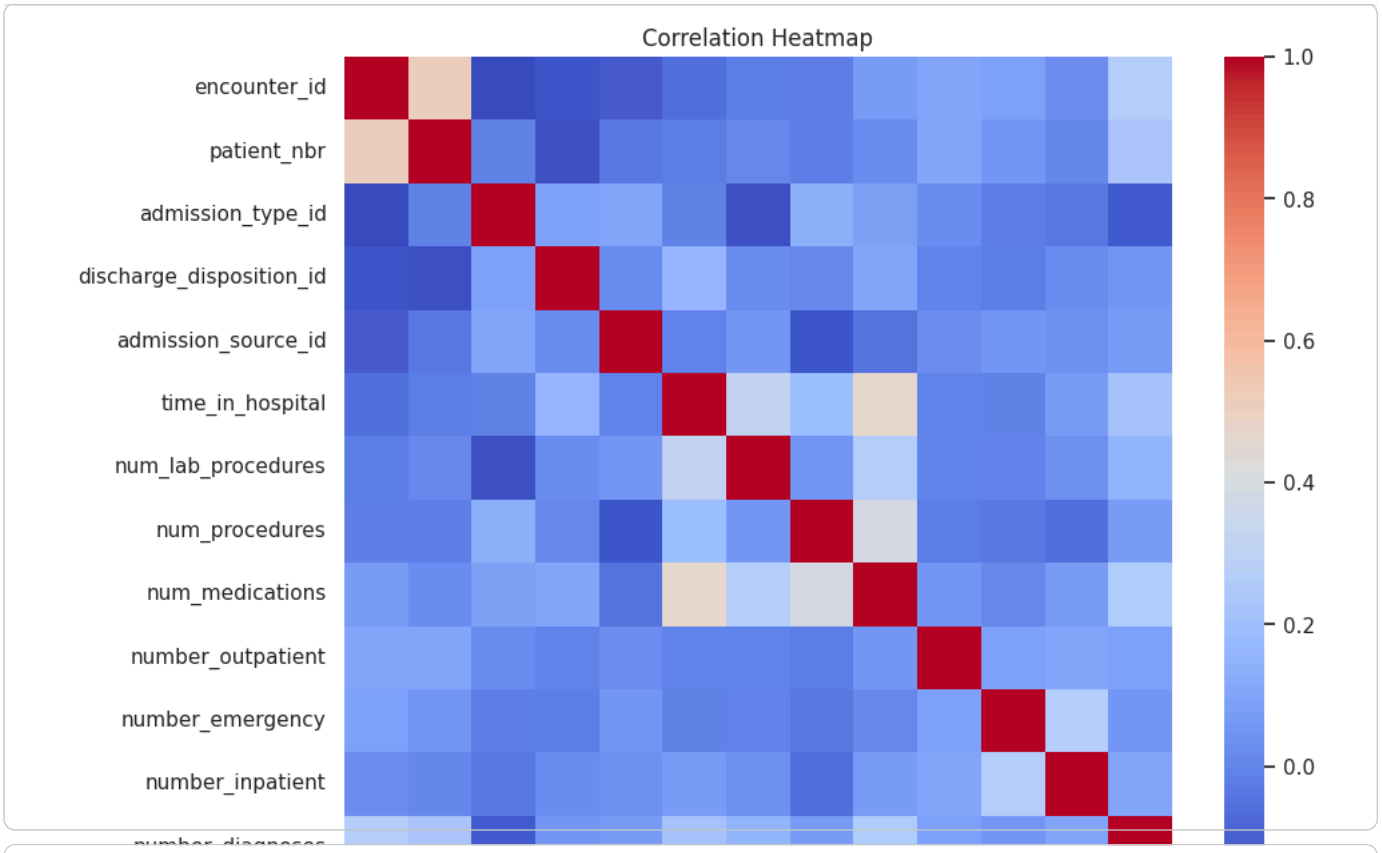
```
sns.histplot(df['time_in_hospital'], bins=20)
plt.title("Time in Hospital Distribution")
plt.show()

sns.histplot(df['num_lab_procedures'], bins=20)
plt.title("Lab Procedures Distribution")
plt.show()
```






```
num_cols = df.select_dtypes(include=['int64', 'float64'])
plt.figure(figsize=(10,8))
sns.heatmap(num_cols.corr(), cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```




```
sns.boxplot(x='readmitted', y='time_in_hospital', data=df)
plt.title("Readmission vs Time in Hospital")
plt.show()
```